

Project Report

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Heart Disease Prediction Using Machine Learning

1. Introduction

Heart disease is one of the leading causes of death worldwide. Early detection plays a crucial role in improving patient outcomes and reducing risks. In this project, machine learning techniques are applied to predict the presence of heart disease based on clinical and medical attributes. The objective is to analyze patient data and build predictive models that can support early diagnosis.

2. Dataset Description

The dataset used in this project is a heart disease dataset -from Kaggle website- containing 297 patient records and 14 attributes. The target variable is **condition**, which indicates whether a patient has heart disease (1) or not (0). The features include medical measurements such as age, cholesterol level, resting blood pressure, maximum heart rate, and other clinical indicators.

3. Data Preprocessing

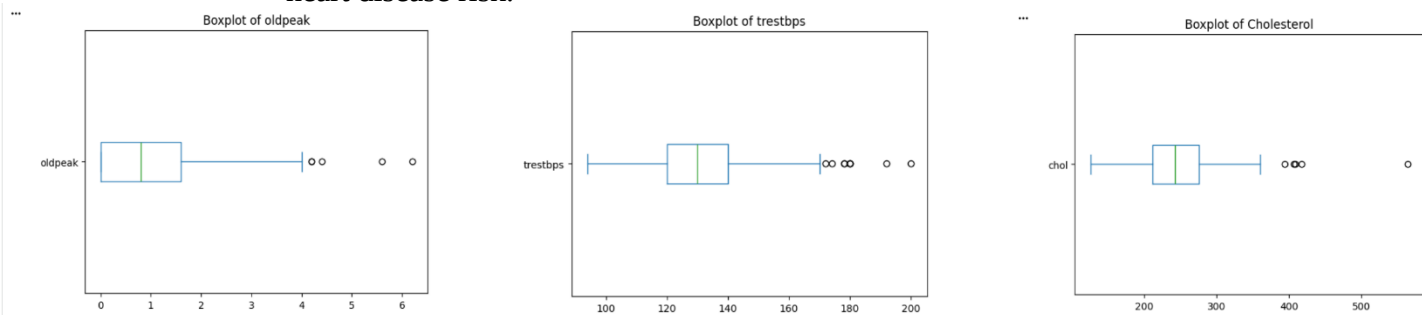
The dataset was carefully examined before model training.

- No missing values were detected.
- No duplicate records were found.
- Data types were consistent and appropriate.
- Categorical variables such as cp, thal, and slope were encoded using one-hot encoding.
- Feature scaling was applied for KNN and Logistic Regression models to ensure fair distance and weight calculations.

4. Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to understand feature distributions and relationships.

- Boxplots revealed the presence of outliers in features such as cholesterol, resting blood pressure, and ST depression (oldpeak).
- These outliers were considered clinically meaningful rather than errors, as they may represent high-risk patients.
- Correlation analysis showed that certain features are strongly associated with heart disease risk.



5. Machine Learning Models

Three classification models were implemented:

- Decision Tree Classifier
- K-Nearest Neighbors (KNN)
- Logistic Regression

Each model was trained on the training dataset and evaluated on unseen test data.

6. Model Evaluation

The models were evaluated using accuracy score, confusion matrix, and classification report.

- Decision Tree Accuracy: 64%
- KNN Accuracy: 80%
- Logistic Regression Accuracy: 84%
- Logistic Regression achieved the highest performance among all models, and this is the confusion matrix:

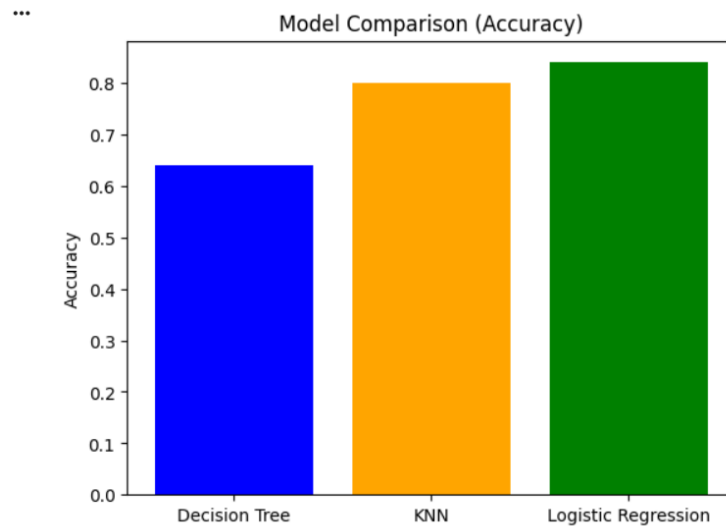
	Predicted 0	Predicted 1
Actual 0	30	4
Actual 1	8	33

- 30 patients were correctly classified as healthy (True Negatives)
- 33 patients were correctly classified as having heart disease (True Positives)
- 4 healthy patients were incorrectly classified as sick (False Positives)
- 8 sick patients were incorrectly classified as healthy (False Negatives)

7. Model Comparison

A comparison of all models shows that:

- Decision Tree is easy to interpret but less accurate and more sensitive to data variations.
- KNN performs well after feature scaling but depends heavily on distance calculations.
- Logistic Regression provides the best balance between accuracy and stability for this dataset.



8. Key Insights

- Medical features such as cholesterol level, maximum heart rate, and ST depression are strong predictors of heart disease.
- Feature scaling significantly improves performance for distance-based models like KNN.
- Model comparison helps identify the most suitable algorithm for this dataset.

10. Conclusion

This project demonstrates how machine learning can be used to analyze medical data and predict heart disease risk. Among the tested models, Logistic Regression performed best with an accuracy of 84%, making it the most suitable model for this dataset.

The results highlight the potential of machine learning in supporting early detection of heart disease. However, further improvements such as larger datasets, feature engineering, and advanced algorithms could enhance prediction performance and reliability.